

RAM and Storage

1. Open your own laptop or desktop (or use a lab machine) and use CPU-Z to find out the exact RAM type (DDR3, DDR4, or DDR5), frequency (in MHz), and total installed memory. Take a screenshot of the Memory tab and note down these details.

->You can complete this practical task on your own laptop/desktop using **CPU-Z**. Since I cannot access your computer directly, use these steps to find the exact values.

Steps to Check RAM Using CPU-Z

1. Download and install **CPU-Z** on your computer.
2. Open **CPU-Z**.
3. Click the **Memory** tab.
4. Note these details:
 - **Type:** DDR3 / DDR4 / DDR5
 - **DRAM Frequency:** shown in MHz
 - **Total Installed Memory:** Check the **General** section or the **SPD** tab/Windows Task Manager for total RAM.
5. Take a **screenshot of the Memory tab** showing the RAM information.
6. Add the screenshot to your assignment.

Assignment Table

RAM Detail	Your Computer
RAM Type	_____
Frequency (MHz)	_____
Total Installed Memory	_____
CPU-Z Memory Tab Screenshot	Attach screenshot here

Important: CPU-Z's **DRAM Frequency** may appear lower than the RAM's advertised effective speed because DDR memory transfers data twice per clock cycle. For example, around **1600 MHz DRAM Frequency corresponds to DDR4-3200**.

2. Physically examine a RAM module and a storage drive (HDD or SSD) available to you. Identify and write down the form factor (e.g., DIMM, SO-DIMM for RAM; 2.5-inch, M.2 for storage) and the connector type (e.g., SATA, M.2, PCIe) for each.

->You can use the following format for your practical assignment. Check your actual RAM and storage drive and replace the blanks with what you observe.

Physical Examination of RAM and Storage Drive

Component	Form Factor	Connector Type	Identification
RAM Module	DIMM / SO-DIMM	DDR4 / DDR5 memory slot	Desktop RAM is usually DIMM; laptop RAM is usually SO-DIMM
Storage Drive	2.5-inch / M.2	SATA / PCIe (NVMe)	2.5-inch drives commonly use SATA; M.2 SSDs may use SATA or PCIe/NVMe

Example Observation

If you are examining a typical **laptop**, your answer could look like:

- **RAM:** SO-DIMM, DDR4, connected through a laptop SO-DIMM memory slot.
- **Storage:** M.2 SSD, PCIe/NVMe interface, connected through an M.2 slot.

If you are examining a typical **desktop PC**:

- **RAM:** DIMM, DDR4/DDR5, connected through a DIMM slot.
- **Storage:** 2.5-inch SATA SSD or HDD, using a SATA data connector and SATA power connector.

Tip: Take a clear photo of the RAM module and storage drive and attach it to your assignment as proof of physical examination.

3. Create a comparison table with at least 4 points contrasting HDD and SSD based on speed, durability, cost, and typical use cases. For each point, give a real-world example (e.g., 'SSD: Used in gaming laptops for fast load times').

->HDD vs SSD Comparison

Comparison Point	HDD (Hard Disk Drive)	SSD (Solid State Drive)	Real-World Example
Speed	Slower because it uses spinning magnetic disks and a moving read/write head.	Much faster because it uses flash memory and has no moving parts.	SSD: Used in gaming laptops for faster game and application loading.
Durability	More vulnerable to damage from physical shocks because it has moving parts.	Generally more resistant to physical shocks because it has no moving mechanical parts.	SSD: Suitable for laptops that are frequently moved around.
Cost	Usually cheaper per GB, especially for large capacities.	Usually costs more per GB than an HDD.	HDD: Used for affordable 2–8 TB desktop storage and backups.
Typical Use Cases	Good for large-capacity storage, backups, and storing photos/videos.	Good for operating systems, gaming, applications, and fast everyday computing.	SSD: Used as the main drive in modern PCs for quick startup and responsiveness.
Noise & Power	Can produce some noise and generally uses more power because of its moving parts.	Silent operation and generally lower power consumption.	SSD: Useful in laptops where quiet operation and battery efficiency are important.

Conclusion

HDDs are a good choice when you need **large storage at a lower cost**, while **SSDs** are preferred when you need **fast performance, quick loading, and better resistance to physical movement**.

4. Calculate the minimum storage capacity needed for each of these scenarios: (a) a YouTuber editing 4K videos weekly, (b) a programmer with lots of code projects and IDEs, (c) a gamer who wants to install 5 AAA games (assume each game is 100GB). Show your calculations

-> Here are the calculations. For scenarios (a) and (b), I'll use reasonable minimum storage assumptions because the question does not specify file sizes.

Minimum Storage Capacity Calculation

Scenario	Calculation	Minimum Storage Needed
(a) YouTuber editing 4K videos weekly	4K video footage ≈ 100 GB/week $\times 4$ weeks = 400 GB	400 GB
(b) Programmer with many code projects and IDEs	Code/projects ≈ 100 GB + IDEs/tools ≈ 100 GB + OS/other files ≈ 100 GB	300 GB
(c) Gamer installing 5 AAA games	5 games $\times 100$ GB = 500 GB	500 GB

Final Answer

(a) YouTuber:

$100 \text{ GB} \times 4 \text{ weeks} = \mathbf{400 \text{ GB}}$

(b) Programmer:

$100 \text{ GB} + 100 \text{ GB} + 100 \text{ GB} = \mathbf{300 \text{ GB}}$

(c) Gamer:

$5 \times 100 \text{ GB} = \mathbf{500 \text{ GB}}$

Therefore, based on these assumptions, the minimum capacities are **400 GB, 300 GB, and 500 GB**, respectively. In practice, a larger drive is recommended so there is room for the operating system, updates, temporary files, and future data.

5. Based on your research and calculations, recommend the best storage solution (type and interface) for these three users: (1) a college student using only MS Office and web browsing, (2) a mobile app developer running Android Studio, (3) a competitive gamer who streams on Twitch. Justify your choice for each.

-> Based on the storage calculations and current technical guidance, here is a suitable recommendation for each user:

Storage Recommendations

User	Recommended Storage	Interface	Why?
1. College student – MS Office & web browsing	500 GB SATA SSD	SATA III	Office applications, web browsing, and everyday tasks do not require extremely high storage speeds. A SATA SSD provides fast booting and application loading at a reasonable cost. SATA SSDs are considered suitable for general office and web-browsing workloads. (Sandisk)
2. Mobile app developer – Android Studio	1 TB NVMe SSD	PCIe/NVMe	Android Studio involves IDE files, SDKs, project files, Gradle builds, and potentially emulators. Google recommends an SSD with at least 32 GB of free space for Android Studio and the emulator. NVMe provides higher throughput and lower response times than SATA SSDs, making it well suited to development workloads. (Android Developers)
3. Competitive gamer who streams on Twitch	1 TB–2 TB NVMe SSD	PCIe/NVMe	Games can occupy substantial storage, while streaming creates additional files such as recordings and clips. NVMe provides fast loading and high storage performance. However, SSD speed does not directly increase game FPS; CPU and GPU performance are more important for FPS. (Sandisk)

Justification

1. College Student:

A **500 GB SATA SSD** is sufficient for MS Office, web browsing, documents, and normal applications. An SSD provides faster startup and application loading than an HDD, while SATA performance is already adequate for these light workloads. ([Sandisk](#))

2. Mobile App Developer:

A **1 TB NVMe SSD** is recommended because development environments can involve Android Studio, SDKs, emulators, Gradle caches, and multiple projects. Android Studio's official requirements specifically recommend an SSD with 32 GB or more of free space for Studio and the emulator. ([Android Developers](#))

3. Competitive Gamer and Streamer:

A **1–2 TB NVMe SSD** provides enough room for several large games and recorded content while offering fast loading and file-transfer performance. NVMe is particularly useful for demanding storage workloads, although it should not be confused with a direct FPS upgrade. ([Sandisk](#))

Overall Recommendation

- **Basic student:** 500 GB SATA SSD
- **Android developer:** 1 TB NVMe SSD
- **Competitive gamer/streamer:** 1–2 TB NVMe SSD

These recommendations balance **capacity, speed, and the workload of each user** rather than choosing the same storage type for everyone.

The exact capacity can be adjusted depending on the amount of games, projects, videos, and other files each person stores.